



KDA-Tuning: Knowledge-Decoupled Adapter Tuning for Vision-Language Models

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Abstract. Pre-trained vision-language models (VLMs) have achieved remarkable performance across various downstream tasks. To further enhance the performance of VLMs on downstream tasks, adapter tuning has emerged as an efficient approach, enabling effective adaptation with minimal parameter updates. However, existing methods are susceptible to overfitting on task-specific data distributions, resulting in suboptimal performance. To address this challenge, we propose a novel framework, knowledge-decoupled adapter tuning (KDA-Tuning), which efficiently adapts VLMs to downstream tasks while preserving generalizability. KDA-Tuning employs a knowledge-decoupled adapter with a dual-branch structure consisting of one branch for general knowledge and another for task-specific knowledge, avoiding conflicts between their optimization. Additionally, we introduce corresponding supervision, including the general knowledge retention loss to encourage the models to preserve general knowledge, and the task-specific knowledge constraint loss to enhance cross-modal alignment for distinguishable feature representations. Extensive experiments demonstrate that KDA-Tuning outperforms existing methods on typical benchmarks.

Keywords: Knowledge-decoupled · Adapter Tuning · Vision-language Models

1 Introduction

Pre-trained vision-language models (VLMs) [1–4] have demonstrated remarkable generalization capabilities. To further enhance their performance on downstream tasks, recent studies focus on efficient tuning paradigms, primarily categorized into prompt tuning [5–10] adapter tuning [11, 12]. Despite the great progress in efficient tuning, existing methods still face several challenges. Methods like CoOp [5] and DAPT [9] face an optimization conflict: overfitting to downstream data distributions compromises the model's ability to generalize to unseen tasks. Moreover, methods such as CLIP-Adapter [11] rely on rigid residual connections with fixed fusion ratios, necessitating extensive experiments for optimal settings across tasks, thus limiting flexibility.

To address these challenges, we propose knowledge-decoupled adapter tuning (KDA-Tuning), a novel framework that explicitly decouples general and task-specific knowledge in adaptation of VLMs. Our key insight is that latent VLM representations

contain both general semantic knowledge and task-discriminative features. Direct optimization of these intertwined components creates conflicting objectives, leading to either over-specialization or under-adaptation. By decoupling these two components, KDA-Tuning enables models to retain general knowledge while effectively adapting to task-specific data distributions. KDA-Tuning employs a dual-branch knowledge-decoupled adapter consisting of a general branch and a task-specific branch, as shown in Fig. 1(a). The general branch preserves the inherent generalization ability of pre-trained VLMs, while the task-specific branch specializes in adapting to downstream tasks. A dynamic fusion module integrates the outputs of these branches through a learnable linear layer, allowing the model to flexibly determine the optimal adaptation pathway.

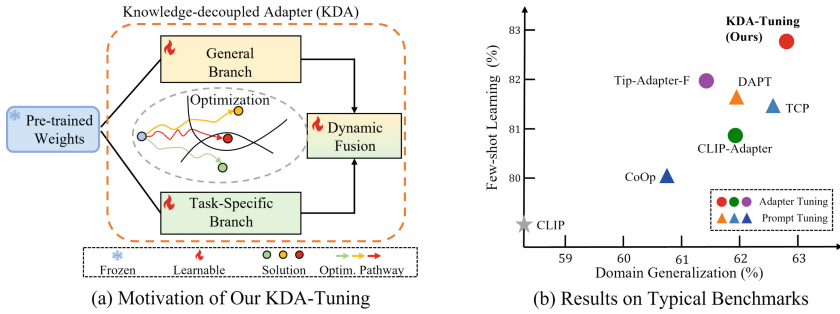


Fig. 1. Motivation and performance of KDA-Tuning.

To facilitate KDA-Tuning, we introduce corresponding supervision. (i) General knowledge retention loss, which employs knowledge self-distillation to encourage the general branch to mimic the frozen pre-trained model, thereby ensuring the preservation of general knowledge. (ii) Task-specific knowledge constraint loss, which enforces alignment between visual and textual embeddings within the task-specific branch, improving modality interaction and representation discrimination, particularly in low-data settings. By integrating these losses with the standard cross-entropy loss, KDA-Tuning enables effective knowledge transfer while maintaining the generalization capabilities of pre-trained VLMs. As shown in Fig. 1(b), KDA-Tuning achieves superior performance on various benchmarks, highlighting its effectiveness in balancing generalization and task adaptation. In summary, our contributions are as follows:

- We propose KDA-Tuning, a novel tuning framework for VLMs that explicitly decouples task-specific and general knowledge, improving the adaptability of VLMs.
- We design a knowledge-decoupled adapter with a dual-branch structure, effectively mitigating conflicts between general and task-specific optimization and automatically optimizes the combination ratio between branches.
- We introduce corresponding supervision to encourage models to focus on general knowledge and task-specific knowledge during KDA-Tuning.
- We conduct extensive experiments, demonstrating that KDA-Tuning achieves superior performance on few-shot learning and domain generalization tasks.

2 Related Work

Recent advances in VLMs, exemplified by CLIP [1], ALIGN [2], FLIP [3] and Florence [4] have revolutionized multimodal representation learning through large-scale pretraining on image-text pairs. These models learn universal semantic alignments that enable strong zero-shot transfer across diverse vision-language tasks. Traditional full-parameter tuning becomes impractical for large VLMs due to computational costs and overfitting risks in few-shot settings. Recent efforts explore parameter-efficient tuning through two primary paradigms, prompt tuning and adapter tuning. Prompt tuning injects learnable text prompts into the encoder to guide task adaptation. CoOp [5] pioneers continuous prompt optimization for CLIP, while CoCoOp [6] enhances instance adaptability via image-conditioned prompts. Subsequent works [7–10] improve prompt design. However, prompt tuning requires gradient propagation through the entire text encoder, incurring substantial computational overhead. Adapter tuning attaches lightweight modules to frozen VLM encoders. CLIP-Adapter [11] introduces residual MLPs to refine visual features, and Tip-Adapter [12] augments CLIP with a cached feature bank for nearest-neighbor classification.

3 Knowledge-Decoupled Adapter Tuning

In this section, we introduce our KDA-Tuning, which consists of knowledge-decoupled adapters (KDA) and corresponding supervision, as illustrated in Fig. 2.

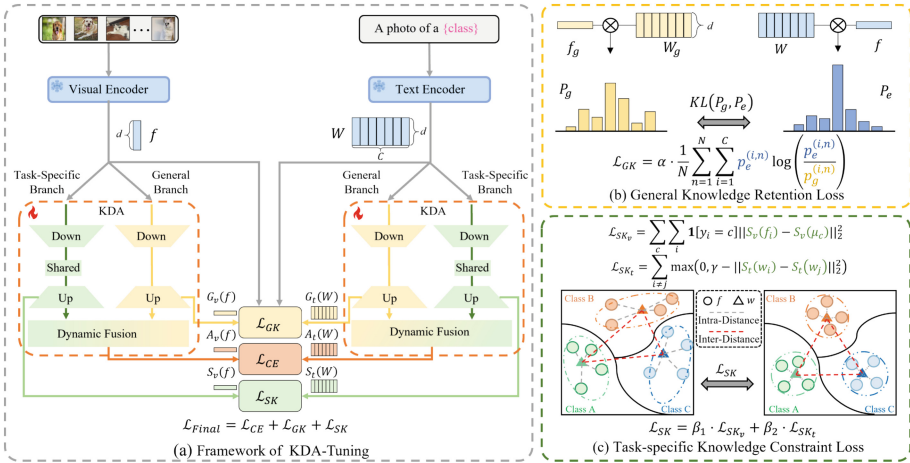


Fig. 2. Overview of KDA-Tuning

3.1 Preliminaries

CLIP [1] is a vision-language model trained with contrastive learning. CLIP comprises two main components: a visual encoder E_v and a textual encoder E_t . Given an image

x and handcrafted textual descriptions of C classes, denoted as $\{t_i\}_{i=1}^C$, CLIP extracts D -dimensional feature representations for both modalities. The visual representation is denoted as $f \in R^D$, while the textual representations, denoted as a set of class embeddings $\{w_i\}_{i=1}^C$, are organized into a matrix $W \in R^{C \times D}$. The encoding process can be formalized as:

$$f = E_v(x), \quad w_i = E_t(t_i), \quad (1)$$

where w_i represents the textual embedding corresponding to the i -th class. Both the visual embedding f and textual embeddings $\{w_i\}_{i=1}^C$ are normalized to unit length, ensuring effective alignment between the two modalities and facilitating the computation of semantic similarities. Furthermore, prediction probability for the i -th class is computed using the softmax function, which operates on cosine similarity between visual embedding f and textual embeddings $\{w_i\}_{i=1}^C$. This is expressed as:

$$p_i = \frac{\exp(f \cdot w_i^\top / \tau)}{\sum_{j=1}^C \exp(f \cdot w_j^\top / \tau)} \quad (2)$$

where τ is a temperature parameter that controls the sharpness of probability distribution. By incorporating cross-entropy loss, CLIP learns to align the visual and textual representations, ensuring that the predicted probabilities correspond accurately to the correct labels. This process can be formalized as:

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{n=1}^N \sum_{i=1}^C \mathbf{1}[y^n = i] \log(p_i^n), \quad (3)$$

where N denotes the number of training samples, y^n is the true label of the n -th sample, and p_i^n is the predicted probability for the i -th class for the n -th sample. The indicator function $\mathbf{1}[y^n = i]$ equals to 1 if y^n corresponds to i -th class, and 0 otherwise.

3.2 Knowledge-Decoupled Adapter

Knowledge-decoupled adapters are the fundamental component of KDA-Tuning, consisting of a dual-branch structure with a general branch and a task-specific branch. Knowledge-decoupled adapters explicitly decouple knowledge with independent branches to avoid conflicts during the optimization of general and task-specific representations. Then, knowledge is dynamically fused to perform classification predictions, enabling the model to adaptively determine the optimal optimization pathway.

In practice, we integrate two knowledge-decoupled adapters, $A_v(\cdot)$ and $A_t(\cdot)$, into the visual and textual branches, which share a similar structure, as shown in Fig. 2(a). Taking the visual adapter $A_v(\cdot)$ as an example, it consists of a general branch $G_v(\cdot)$ and a task-specific branch $S_v(\cdot)$, both implemented as lightweight multi-layer perceptrons (MLPs). During downstream task adaptation, the encoded features are processed through both branches and dynamically fused via the fusion module W_v^o , a single linear layer. Notably, to mitigate the semantic gap between two modalities, particularly in scenarios with limited training samples, a shared projection layer $W^{(s,r)}$ is incorporated into the task-specific branches of $A_v(\cdot)$ and $A_t(\cdot)$. This layer enables gradient propagation interaction

between the two modalities, thereby facilitating cross-modal semantic alignment. To simplify, we adopt a unified formulation for $A_v(\cdot)$ and $A_t(\cdot)$, denoted as $A_{v,t}(\cdot)$. Similarly, the components $G_v(\cdot)$, $G_t(\cdot)$, $S_v(\cdot)$, and $S_t(\cdot)$ are unified as $G_{v,t}(\cdot)$ and $S_{v,t}(\cdot)$, respectively. These modules are defined as follows:

$$G_{v,t}(z) = \delta\left(W_{v,t}^{(g,u)} \cdot \delta\left(W_{v,t}^{(g,d)} \cdot z\right)\right), \quad (4)$$

$$S_{v,t}(z) = \delta\left(W_{v,t}^{(s,u)} \cdot \delta\left(W^{(s,r)} \cdot \delta\left(W_{v,t}^{(s,d)} \cdot z\right)\right)\right), \quad (5)$$

$$A_{v,t}(z) = \delta\left(W_{v,t}^o \cdot (G_{v,t}(z) \oplus S_{v,t}(z))\right), \quad (6)$$

where z denotes the D -dimensional input of the adapters, which can either be the visual embedding f or textual embeddings $\{w_i\}_{i=1}^C$, both encoded by the pre-trained CLIP and obtained from Eq. 1. Let D_q denote $D/4$. The general branch utilizes up-projection $W_{v,t}^{(g,u)} \in \mathbb{R}^{D \times D_q}$ and down-projection $W_{v,t}^{(g,d)} \in \mathbb{R}^{D_q \times D}$, which encode general semantic features. Similarly, the task-specific branch employs $W_{v,t}^{(s,u)} \in \mathbb{R}^{D \times D_q}$ and $W_{v,t}^{(s,d)} \in \mathbb{R}^{D_q \times D}$, along with an additional shared projection layer $W^{(s,r)} \in \mathbb{R}^{D_q \times D_q}$ for cross-modal semantic alignment, to capture task-specific representations. The dynamic fusion layer $W_{v,t}^o \in \mathbb{R}^{D \times 2D}$ integrates the outputs of the general and task-specific branches via concatenation $\oplus$. And $\delta(\cdot)$ denotes a non-linear activation function. By leveraging Eq. (6), the adapted visual feature f^* and textual embeddings W^* are derived as follows:

$$f^* = A_v(f), \quad W^* = A_t(W). \quad (7)$$

With the final visual feature f^* and textual embeddings W^* , the category probability predictions $P = \{p_i\}_{i=1}^C$, are subsequently computed using Eq. 2.

3.3 Supervision for KDA-Tuning

We introduce corresponding supervision. Specifically, for the final prediction results $P = \{p_i\}_{i=1}^C$, we substitute them into Eq. 3 to compute $\mathcal{L}_{CE}$. We adopt cross-entropy loss $\mathcal{L}_{CE}$ to optimize the adapters $A_v(\cdot)$ and $A_t(\cdot)$ globally during tuning, enabling them to adapt to downstream tasks. Considering the characteristics of our knowledge-decoupled adapters, we introduce two auxiliary losses to encourage the two branches to focus on their respective knowledge. The former is general knowledge retention loss $\mathcal{L}_{GK}$, which ensures retention of the general knowledge for generalizability. The latter is task-specific knowledge constraint loss $\mathcal{L}_{SK}$, which enhances task-specific representations. Combining these losses, $\mathcal{L}_{Final}$ is defined as:

$$\mathcal{L}_{Final} = \mathcal{L}_{CE} + \mathcal{L}_{GK} + \mathcal{L}_{SK}. \quad (8)$$

General Knowledge Retention Loss. Pre-trained CLIP, trained on large-scale image-text data, encapsulates universal representations. Therefore, we utilize the frozen pre-trained CLIP as the source of general knowledge and supervise the learning of the general branch through knowledge distillation, as shown in Fig. 2(b). Specifically, we employ the visual and textual features encoded by pre-trained CLIP, denoted as f and W , which

are obtained from Eq. 1. Meanwhile, the corresponding features from the general branch, f_g and W_g , are computed as:

$$f_g = G_v(f), \quad W_g = G_t(W). \quad (9)$$

Then, we substitute them into Eq. 2 to derive the prediction distribution of the frozen CLIP, $\{p_e^i\}_{i=1}^C$, denoted as P_e , and the prediction distribution representing general knowledge, $\{p_g^i\}_{i=1}^C$, denoted as P_g . Then we introduce the general knowledge retention loss, which forces the general branch to mimic frozen pre-trained CLIP, and is implemented by minimizing the Kullback-Leibler (KL) divergence between the two prediction distributions. The general knowledge retention loss, $\mathcal{L}_{GK}$, is formulated as:

$$\mathcal{L}_{GK} = \alpha \cdot \frac{1}{N} \sum_{n=1}^N \sum_{i=1}^C p_e^{(i,n)} \log \left(\frac{p_e^{(i,n)}}{p_g^{(i,n)}} \right), \quad (10)$$

where N denotes the number of training samples, and C represents the number of classes. $p_e^{(i,n)}$ and $p_g^{(i,n)}$ represent the probabilities of the i -th class for the n -th sample, generated by the frozen pre-trained CLIP and the general branch, respectively. α is a hyperparameter that controls the weight.

Task-Specific Knowledge Constraint Loss. VLMs align visual and textual representations of the same class in the latent semantic space via contrastive learning. After training with large-scale image-text pairs, textual embeddings can be approximated as the centers of corresponding visual features in the embedding space, effectively capturing category comprehensive semantic information. However, due to the limited training samples in downstream tasks, the significant semantic gap between the visual and textual modalities hinders cross-modal alignment, limiting the performance of CLIP. To address this issue, we impose explicit constraints on visual and textual embeddings within the task-specific knowledge branch, enhancing cross-modal alignment for distinguishable feature representations. So we propose the task-specific knowledge constraint loss, as illustrated in Fig. 2(c). Specifically, we first propose the visual task-specific loss $\mathcal{L}_{SK_v}$, which aims to bring visual embeddings of the same class closer to their corresponding class prototype. Meanwhile, for textual embeddings, we introduce the textual task-specific loss $\mathcal{L}_{SK_t}$, which constrains the distance between textual embeddings to prevent overly small inter-class distances, thereby enhancing task-specific representations. The corresponding loss functions are defined as follows:

$$\mathcal{L}_{SK_v} = \sum_{c=1}^C \sum_{n=1}^N \mathbf{1}[y_n = c] \|S_v(f_n) - S_v(\mu_c)\|_2^2, \quad (11)$$

$$\mathcal{L}_{SK_t} = \sum_{i \neq j} \max(0, \gamma - \|S_t(w_i) - S_t(w_j)\|_2^2), \quad (12)$$

where $S_v(\cdot)$ and $S_t(\cdot)$ represent task-specific transformations for visual and textual embeddings, as defined in Eq. (5). C denotes the number of classes, and N denotes the number of training samples. f_n denotes visual embedding of the n -th training samples. w_i and w_j represent the textual embeddings of the i -th and j -th classes, respectively. f_n , w_i , and w_j are extracted from the frozen CLIP encoder, as detailed in Eq. (1). The

indicator function $\mathbf{1}[y_n = c]$ ensures that only samples belonging to class c contribute to the loss. The class prototype μ_c is computed as:

$$\mu_c = \frac{1}{|D_N^c|} \sum_{(x_i, y_i) \in D_N^c} E_v(x_i), \quad (13)$$

where $D_N = \{(x_i, y_i)\}_{i=1}^N$, represents the training dataset containing N samples. x_i denotes an image in D_N , and y_i is its corresponding label. The subset $D_N^c = \{(x_i, y_i) \in D_N \mid y_i = c\}$ consists of samples belonging to class c , with $|D_N^c|$ indicating the size of D_N^c . Due to textual embeddings representing the comprehensive semantics of the category information, the parameter γ , which defines the separation margin between different classes, is computed as:

$$\gamma = \eta \cdot \frac{1}{C(C-1)} \sum_{i \neq j} \|w_i - w_j\|_2, \quad (14)$$

where η controls the margin tightness, which is typically set to 2 in practice. Finally, the task-specific knowledge constraint loss $\mathcal{L}_{SK}$ is defined as:

$$\mathcal{L}_{SK} = \beta_1 \cdot \mathcal{L}_{SK_v} + \beta_2 \cdot \mathcal{L}_{SK_t}. \quad (15)$$

Here, β_1 and β_2 are hyperparameters that control loss weights.

4 Experiments

4.1 Experimental Setup

Table 1. Domain generalization performance of KDA-Tuning and previous SOTA methods.

Method	Source	Target				Average
	ImageNet	-V2	-S	-A	-R	
CLIP	66.67	60.88	46.11	47.75	73.97	59.08
CoOp	71.93	64.22	47.07	47.97	74.32	60.83
CLIP-Adapter	72.11	62.36	47.72	46.20	75.81	61.99
TIP-Adapter-F	72.33	63.07	48.75	47.23	76.18	61.51
Maple	70.62	63.97	49.05	49.82	76.23	62.27
DAPT	72.50	64.93	48.30	48.74	75.75	62.06
TCP	72.28	64.50	49.12	49.69	76.53	62.42
KDA-Tuning	72.92	65.16	49.68	49.66	76.51	62.79

Datasets and Benchmarks. We comprehensively evaluate KDA-Tuning across two critical benchmarks: few-shot learning and domain generalization. For few-shot learning, we follow standard protocols, conducting [1, 2, 4, 8, 16]-shot experiments on 11 diverse downstream datasets, including ImageNet, StanfordCars, Caltech101, UCF101, Flowers102, Food101, DTD, EuroSAT, FGVCaircraft, OxfordPets, and SUN397. This benchmark evaluates model adaptability under limited supervision by training on K-shot samples per class and testing on full test sets. To evaluate domain generalization, ImageNet serves as the source domain, with testing conducted on four distribution-shifted datasets: ImageNet-V2, -S, -A, and -R. This benchmark examines the model’s resilience to domain changes while ensuring category-level semantic alignment.

Comparison Methods. We compare our KDA-Tuning with several state-of-the-art (SOTA) methods, including Zero-shot CLIP [1], CoOp [5], CLIP-Adapter [11], DAPT [9], Maple [7], TIP-Adapter-F [12], and TCP [10]. To ensure fair comparisons, we follow the same settings as the original implementations, including feature backbone, prompt length, and learning rate. For Zero-shot CLIP, we evaluate the pre-trained CLIP directly without additional training. CLIP-Adapter and TIP-Adapter-F are implemented with the ViT-B/16 visual encoder, configured according to official settings.

Implementation Details. We utilize the ViT-B/16 [14] and transformer [10] as visual and textual encoder of CLIP respectively, which outputs features with $D = 1024$ channels. Margin tightness η is set to 2. All experiments are repeated three times on NVIDIA RTX 3090 GPUs, and the average results are reported.

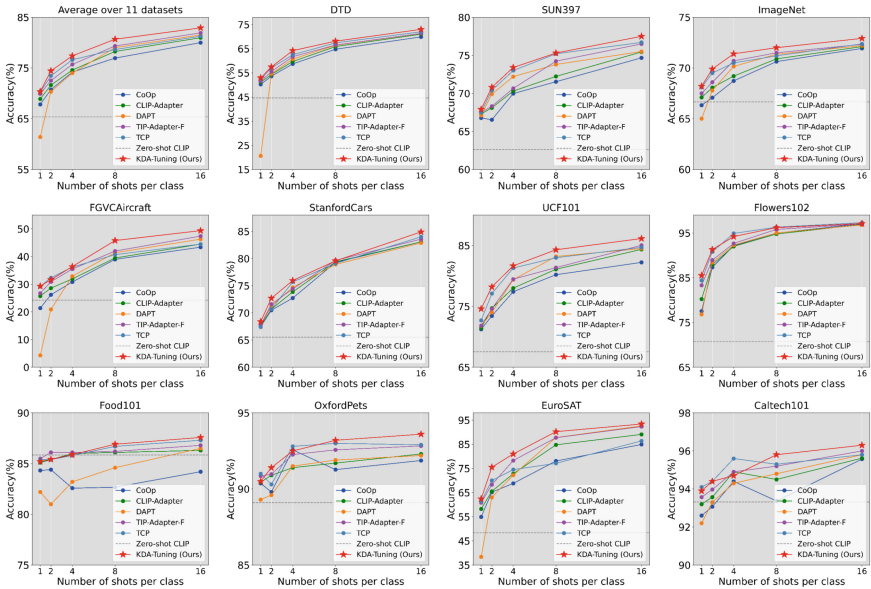


Fig. 3. Few-shot learning performance of KDA-Tuning and previous SOTA methods.

4.2 Comparison with SOTA Methods

Few-Shot Learning. As shown in Fig. 3, KDA-Tuning achieves superior few-shot performance across 11 datasets under [1, 2, 4, 8, 16]-shot settings, reaching SOTA average accuracy of [70.36%, 74.43%, 77.39%, 80.67%, 82.91%]. Take representative settings with very few samples (1-shot), moderate samples (4-shot), and relatively more samples (16-shot) as examples, KDA-Tuning surpasses existing methods consistently. These results highlight KDA-Tuning’s balanced optimization of task-specific adaptation and generalization preservation, validating its design effectiveness.

Domain Generalization. We conduct 16-shot KDA-Tuning on ImageNet and evaluate the model on 4 sub-datasets. As summarized in Table 1, while KDA-Tuning marginally underperforms state-of-the-art methods Maple and TCP on ImageNet-A (−0.26%) and ImageNet-R (−0.12%), it achieves the highest average accuracy across the benchmark. These results validate that our dual-branch knowledge decoupling design effectively enhances vision-language models’ generalization capabilities, particularly in balancing task-specific adaptation with cross-domain robustness.

Table 2. Comparison of training efficiency and computational cost.

Method	Epoch	GFLOPS	Param	Accuracy
CoOp	200	10.41	0.01M	71.93
CLIP-Adapter	200	0.004	0.52M	72.11
TIP-Adapter-F	50	0.030	16.3M	72.33
Maple	200	17.52	0.2M	70.72
KDA-Tuning	50	0.019	2.1M	72.92

Table 3. Effectiveness of shared projection layer. (w/o: without, w/: with)

Shared Projection	1-shot	4-shot	16-shot
w/o	67.25	70.69	72.57
w/	68.18 (+0.93)	71.36 (+0.67)	72.92 (+0.35)

Training Efficiency and Computational Cost. We compare the computational efficiency in Table 2, reporting training epochs, gradient backpropagation GFLOPs, parameter and accuracy. CoOp incurs high GFLOPs due to full-gradient updates across the text encoder. TIP-Adapter-F requires extensive tuning of cached feature banks, significantly increasing trainable parameters and GFLOPs. In contrast, KDA-Tuning achieves the highest accuracy (72.92%) on ImageNet while maintaining balanced computational efficiency, as its dual-branch adapter operates on frozen backbone features with lightweight linear projections.

4.3 Ablation Studies

Various Adapter Variants. We evaluate three variants of adapters: visual-only, text-only, and joint implementation. For fair comparison, single-branch variants adjust the $\mathcal{L}_{GK}$ and $\mathcal{L}_{SK}$: (i) replacing the dual-branch probability distillation in $\mathcal{L}_{GK}$ with feature distillation from frozen backbones, and (ii) applying $\mathcal{L}_{SK}$ constraints only to their respective modalities. Experiments (Fig. 4) reveal that both visual-only and text-only adapters significantly outperform zero-shot CLIP, with visual adaptation showing greater efficacy. Notably, combining both adapters achieves optimal performance, demonstrating that joint multimodal feature alignment yields complementary benefits.

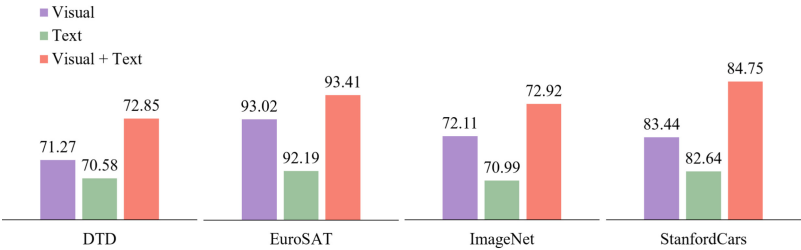


Fig. 4. Impact of various adapter variants.

Table 4. Effectiveness of $\mathcal{L}_{GK}$ and $\mathcal{L}_{SK}$

Method	$\mathcal{L}_{GK}$	$\mathcal{L}_{SK}$	1-shot	4-shot	16-shot
Baseline	×	×	65.64	69.80	72.07
Loss	✓	×	67.35 (+1.71)	70.73 (+0.93)	72.45 (+0.38)
	✓	✓	68.18 (+2.54)	71.36 (+1.56)	72.92 (+0.85)

Effectiveness of Shared Projection Layer. We evaluate the shared projection layer under [1, 4, 16]-shot settings on ImageNet. As shown in Table 3, introducing the shared layer consistently improves performance across all settings, with 0.93% (1-shot), 0.67% (4-shot), and 0.52% (16-shot) accuracy gains. These results highlight critical insights: cross-modal interaction is particularly vital in low-data regimes.

Visualization Analysis. We conduct visualization with Grad-CAM. Results (Fig. 5) reveal distinct attention patterns: the task-specific branch focuses on fine-grained discriminative details (e.g., the car’s faces and rear, the cake’s strawberry) but is susceptible to background noise, while the general knowledge branch prioritizes holistic foreground objects (e.g., vehicle backbone, food center) and overlooks subtle features. Through dynamic fusion, the final prediction synthesizes strengths from both branches.

Effectiveness of $\mathcal{L}_{GK}$ and $\mathcal{L}_{SK}$. We evaluate the contributions of $\mathcal{L}_{GK}$ and $\mathcal{L}_{SK}$ under [1, 4, 16]-shot settings (Table 4). Adding $\mathcal{L}_{GK}$ alone improves accuracy by +1.71% (1-shot), +0.93% (4-shot), and +0.38% (16-shot) over the baseline. Jointly optimizing $\mathcal{L}_{GK}$

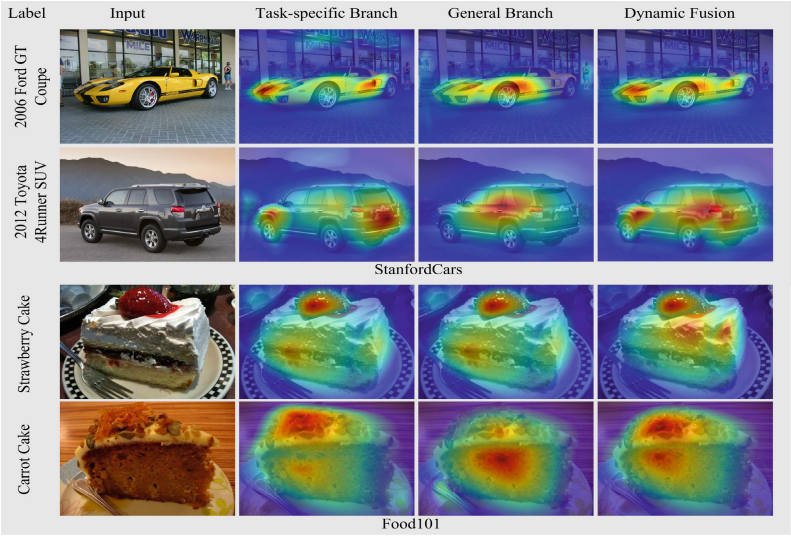


Fig. 5. Grad-CAM results of KDA-Tuning

and $\mathcal{L}_{SK}$ further enhances performance, achieving +2.54% (1-shot), +1.56% (4-shot), and +0.85% (16-shot) improvements, which underscores the complementary nature of hierarchical knowledge constraints: $\mathcal{L}_{GK}$ stabilizes optimization by anchoring pretrained know-ledge, while $\mathcal{L}_{SK}$ refines task-discriminative features.

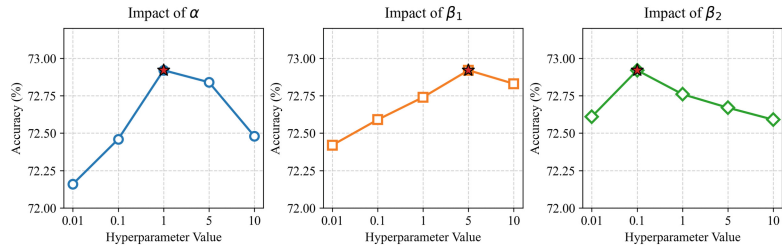


Fig. 6. Hyperparameter α , β_1 , and β_2 sensitivity analysis.

Hyperparameter Sensitivity Analysis. We investigate the impact of hyperparameters α , β_1 , and β_2 (defined in Eq. 10 and Eq. 15) on performance under the 16-shot setting on ImageNet (Fig. 6). Lower α degrade performance, indicating insufficient retention of general knowledge for robust generalization. Optimal β_1 ranges are narrower and more sensitive than those of β_2 . This aligns with CLIP’s inherent properties: text embeddings require minimal downstream adjustment, whereas visual representations demand stronger adaptation due to domain shifts in downstream tasks.

5 Conclusion

In this paper, we propose KDA-Tuning, a novel tuning framework that enhances VLMs adaptation by explicitly decoupling general and task-specific knowledge via dual-branch adapters. By dynamically fusing decoupled representations and enforcing specialized supervision, KDA-Tuning avoids conflicts between the optimization, enabling models to retain generalizability while adapting to task-specific data distributions. Extensive experiments validate the effectiveness of KDA-Tuning in transferring VLMs.

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